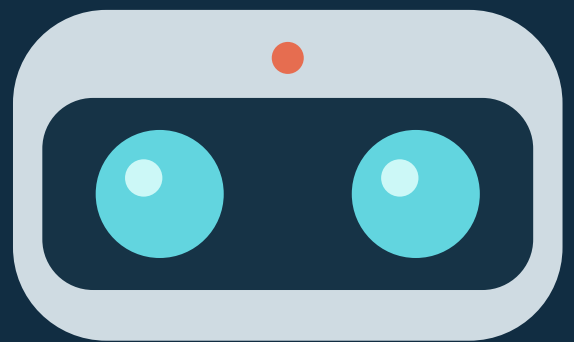


R.O.B. VISION

OPTICAL ROBOT COMPANION / MK.01



USER GUIDE

Play, preview, setup, and problem solving

DESIGN PREVIEW 0.1 / 24 SEPTEMBER 2026

SIMULATION AVAILABLE · PHYSICAL BUILD NOT YET VERIFIED

CHAPTER 01 / DESIGN PREVIEW

R.O.B. Vision user guide

Status: browser preview and local controller trial available; UNO Q game connection remains to be tested. R.O.B. is a virtual character and accessory model. The controller service can watch game flashes through a camera and own the virtual robot state. The robot, gyros, trays, spinner, and Stack-Up discs are drawn in the browser; they are not physical objects.

Try the available preview

Open the [dashboard](#), or serve the project folder locally and open /dashboard/ in a browser. The preview requires no UNO Q, ROM, camera, or network connection. It works on a laptop, iPad, or phone.

The large **Live Model** shows R.O.B. moving two opposing hands around one piece. **Accessory Bay** selects Gyromite, Stack-Up, or Pose Lab. **Game Table** gives a second view of the virtual pieces. **System Vitals** shows virtual pose, gyro spin, and virtual red/blue button state. The smaller game-camera tile is a placeholder. SIMULATION and GAME LINK OFFLINE mean no game is being controlled.

Run Gyromite

1. Select **Gyromite**, then **Run Demo Sequence**.
2. First, watch R.O.B. hold an unspun gyro on the red pad. The virtual red button stays pressed only while he holds it there.
3. Then watch R.O.B. spin Gyro A and B and put them on the red and blue pads. Each gyro has its own illustrative spin-down clock.
4. When each gyro tips, its virtual button releases. R.O.B. re-spins each once. After both stop again, he returns them to their holders and the demo ends.

The 55-second spin time is a demonstration setting, not a historical measurement. Select **Home** to reset at any point, or **Run Demo Sequence** to start again. No player needs to reposition virtual pieces.

Run Stack-Up or Pose Lab

Stack-Up starts with five colored discs on Tray 3. The demo moves red alone to Tray 4, then carries blue and white together to Tray 2, and stops. Use **Stack-Up Commands** below the view to try one virtual movement at a time; the grip button switches between **Close Hands** and **Open Hands**. R.O.B. can lift a disc together with every disc above it. **Home** restores all five discs to Tray 3. **Pose Lab** shows R.O.B. alone: its demo is a short greeting, and the **Pose Preview** buttons let you try turning, lifting, and gripping. It uses no game pieces or optical commands. **Emergency Stop** cancels the local sequence until **Reset Stop** is selected. It is a demo control, not a physical safety switch.

Planned game connection

The cabinet displays Gyromite or Stack-Up on a modern screen. The Uno Q camera will read the game's flashes and validate a complete command before the Uno Q changes the virtual R.O.B. state. A laptop, iPad, or phone will display that state over the LAN. An optional RetroPie launch hook can identify the configured ROM by exact filename; it will not replace flash decoding.

For Gyromite, the UNO Q will send red/blue virtual pad states to a paired virtual Controller 2 receiver over Wi-Fi or Ethernet. A spinning gyro upright on a pad presses it, as does an unspun gyro held there by R.O.B. Moving or tipping the gyro releases it. The game itself sees only Controller 2 input; without an optional emulator hook, R.O.B. Vision cannot know the professor's exact position or prove a game's on-screen response. Stack-Up's documented modes

do not use this tray-button feedback.

For a live controller trial, run `python3 -m controller.service` from the project folder and open `http://127.0.0.1:8766/dashboard/`. Choose a game; commands above System Vitals now go to the controller and survive page reloads. Install `requirements-camera.txt` to enable **Start Camera**. Camera speed and screen decoding must still be validated with the intended hardware. Gyromite cannot yet press buttons in the actual game because its paired Controller 2 receiver is pending. See the [virtual system contract](#), [gameplay guide](#), and [technical architecture](#).

CHAPTER 02 / DESIGN PREVIEW

R.O.B. Vision gameplay guide

Status: scripted preview and local live controller prototype. R.O.B., the gyros, spinner, pads, trays, and discs exist only in the virtual model. The controller service can own their state and update the browser; target UNO Q camera timing and the Gyromite game return path remain unverified.

Gyromite

The original Gyromite booklet describes **Test**, **Direct**, **Game A**, and **Game B**. Direct sends six R.O.B. actions: up, down, left, right, open, and close. In Game A, Start switches into the robot-transmission screen for a command and then back to the professor. In Game B, the professor walks while Controller 1 directly commands R.O.B. A gyro on the blue or red tray lowers the matching gate.

The manual explicitly says a gyro **does not need to spin to operate only one gate**. R.O.B. can hold an unspun gyro on the virtual pad, then lift it to release that pad. If both gates must be controlled, spinning one gyro lets it stay upright on one pad while R.O.B. uses his hands to put the other on the other pad. A gyro that is no longer needed belongs on its holder. The spinner is a means to free R.O.B.'s hands, not an obligatory step for every pad press.

Available preview: the finite Gyromite demo shows an unspun held press, a two-gyro spin relay, separate spin-down and one recovery for each gyro, then a tidy return to both holders. It stops at COMPLETE. The spin lifetime is illustrative. Select **Home** to reset. The browser's virtual button indicators do not yet reach the game.

Planned live controller: the UNO Q maintains the locations and spin phases of Gyro A and B. It derives red and blue virtual button states from those modeled conditions and sends them over LAN to a paired virtual Controller 2 receiver. It releases both buttons on game exit, connection loss, stale packets, or mode change. The exact red/blue-to-Controller-2 A/B mapping remains a bench measurement with the running game. Optical command decoding controls R.O.B.'s virtual actions; a launch hook may supply the game ID. The controller must keep game context and modeled piece state distinct from unknown on-screen professor or gate positions.

Stack-Up

The [Stack-Up](#) booklet describes Direct, Memory, and Bingo. The virtual fixture has five numbered trays and five colored discs. Direct and Memory start with all five on Tray 3, top to bottom **red, white, blue, yellow, green**. One-player Bingo uses any color order on Tray 3; two-player Bingo starts with three on Tray 3 and one each on Trays 2 and 4. Direct moves Professor Hector onto six command keys: **left, right, up, down, open, close**. Memory plays a programmed command series; Bingo sends a command when one row or column completes. The UNO Q will apply each validated optical command as one virtual action. It must model the grippers' height, so R.O.B. can carry a disc together with all discs above it. Stack-Up's documented modes do not need Gyromite's virtual Controller 2 pad return path, even though two-player Bingo uses a second human controller.

Available preview: R.O.B. first moves red alone from Tray 3 to Tray 4, then grips blue and white together and carries both to Tray 2. Green and yellow remain on Tray 3. All five discs appear in the Live Model, Accessory Bay, and Game Table as they move. The **Stack-Up Commands** buttons above System Vitals apply one virtual left, right, up, down, open, or close action. **Home** restores the starting stack. A blocked move leaves all discs in place and explains why. The preview demo is scripted; the separate live controller accepts manual commands and complete ROM-pattern detections from optional camera capture. In game play, the player compares the virtual arrangement with the target shown by the game. The flash stream does not tell the controller which disc to move or whether the target was achieved.

Pose Lab and recovery

Pose Lab exercises R.O.B.'s head, body turn, arms, and paired grip without game pieces or commands. Its demo is a short greeting; the controls below the scene let the viewer try poses manually. It is an art and motion sandbox, not a third game. In a live session, unreadable flashes produce no movement. A browser reconnect receives the UNO Q's current snapshot; it never restarts an action. If the Gyromite return path disconnects, virtual Controller 2 buttons release even if the browser still shows a gyro on a pad. The game host receives controller states, not a proof that a level was completed.

CHAPTER 03 / DESIGN PREVIEW

R.O.B. Vision installation guide

Available now: the local browser preview, exact-name resolver, and Python controller service with optional OpenCV camera capture. **Planned:** validated UNO Q deployment and paired Gyromite game receiver. No physical R.O.B. assembly or motor calibration is needed.

For a local controller trial, run `python3 -m controller.service` from the project folder and open `http://127.0.0.1:8766/dashboard/`. Select a game and try a command; reload to confirm the controller retains state. Install `requirements-camera.txt` to enable **Start Camera**. The secondary camera tile shows a low-rate framing image with the sample region outlined; use `--camera-index` and `--camera-roi x,y,width,height` to start capture with a chosen crop. The reported fps is the delivered rate, not the requested rate. If serving over LAN, set `ROB_VISION_TOKEN` and pass `--host 0.0.0.0`; the dashboard asks for the token. The optional `tools/notify_game.py` helper can send exact RetroPie launch names but is not installed into the cabinet hooks automatically. The existing standalone preview at `dashboard/index.html` stays available without the service.

On the UNO Q, the user service template in `deploy/rob-vision.service` keeps this controller separate from VirtualGlove. Its secret lives in `/home/arduino/.config/rob-vision/environment` with owner-only permissions. It binds port 8766 on the LAN and requires a token for API calls. Camera capture cannot start until a real capture camera appears; `/dev/video0` and `/dev/video1` may be codec devices on UNO Q and are deliberately excluded. Installing OpenCV and checking a camera's delivered fps are separate hardware steps.

Open the preview

From the project directory, serve the files locally with `python3 -m http.server 8000` and open `http://localhost:8000/dashboard/` on the same computer. Select a mode and run its finite demo. **Home** resets the preview; the browser **Emergency Stop** cancels only local scripted actions. The existing preview does not connect to the cabinet or UNO Q.

Planned connected setup

1. Install the supported UNO Q system, camera, and project service after their exact versions and interfaces have been validated.
2. Mount the camera so the cabinet's game flash region stays in frame. Calibrate focus, exposure, and timing using the game's Test mode and capture trace.
3. Connect the UNO Q and game host over trusted Wi-Fi or Ethernet. Pair the Gyromite virtual Controller 2 sender and receiver, then test both button colors and their release on link loss.
4. Serve the dashboard from the UNO Q. Verify laptop, iPad, and phone clients receive a complete snapshot followed by ordered events.
5. Configure exact Gyromite and Stack-Up launch filenames if using the optional RetroPie hook. Verify unknown titles and game exit return to idle.
6. Test a complete optical-command-to-virtual-action loop, then a Gyromite virtual-pad-to-game response loop. Keep ROMs outside the project repository.

Do not invent install commands, UNO Q pin assignments, software versions, or supported camera modes before they are tested. A successful dashboard animation by itself is not proof of game compatibility. See the [verification plan](#).

CHAPTER 04 / DESIGN PREVIEW

R.O.B. Vision troubleshooting

Current preview is local simulation. It does not require an UNO Q, camera, game host, or network pairing.

Symptom	Check
Dashboard does not open	Serve the project directory locally and visit /dashboard/; check the server address and browser console.
Demo appears stuck	Select Home , then Run Demo Sequence . Gyromite includes a long illustrative spin-down, one recovery for each gyro, and holder cleanup before it stops.
Red or blue indicator reads RELEASED	The virtual gyro is off that pad, has tipped, or R.O.B. lifted it. The unspun one-gate example presses only while held down.
Stack-Up shows old Gyromite pieces	Select Stack-Up in Accessory Bay; refresh if a previous page version was cached.
Stop button disables controls	Select Reset Stop ; it resets the local simulation.

Planned UNO Q diagnostics

1. **Game not identified:** check the exact configured ROM basename and launch/exit event. Unknown titles select idle.
2. **No optical command:** inspect camera framing, exposure, timestamped flash trace, and game Test mode. An incomplete message never moves virtual R.O.B.
3. **Virtual R.O.B. does not act:** check accepted command, current mode, busy state, and object-transition rejection log.
4. **Gyromite gate does not respond:** compare virtual red/blue pad state, fresh LAN packet, paired receiver state, A/B mapping, and emulator controller assignment in that order.
5. **Browser shows stale state:** reconnect for a fresh UNO Q snapshot; do not guess position from an old animation.
6. **Return link fails:** both virtual buttons should release; repair pairing or network freshness before continuing.

The UNO Q and game receiver are planned. No connected-system diagnostic commands are implemented yet. Preserve timestamps and event sequences when reporting a future issue, but do not include pairing secrets or ROM files.